

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	I	{Maintain/ control/ prevent changes in} pH (1) keeping pH at optimum/ which would alter the {rate of reaction/ time taken}/ which could denature the enzymes/ prevents coagulation of milk (1)	1	1		2		2
			II	{Maintain/ control/ prevent changes in} temperature/ acclimatise to the same temperature (1) so that all of the enzymes and substrates had the same <u>kinetic</u> energy / if some test tubes were incubated at a higher temperature the enzymes (and substrates) would have more <u>kinetic</u> energy / ORA (1) Ignore references to denaturing/ optimum	1	1		2		2
		(ii)		91.2		1		1	1	1
		(iii)		Correct plots (2) $\pm \frac{1}{2}$ small square 5 correct = 2 marks 4 correct = 1 mark 0/1/2/3 = 0 marks Centre of plots joined with a ruler, no extrapolation/ line of best fit with ruler (allow extrapolation) (1) Range bars correct (1) $\pm \frac{1}{2}$ small square		4		4	4	4